

Foundations Notebook

Your Name

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1 Mathematical Foundations

1.1 Linear Algebra

1.1.1 Vector Spaces

Definition 1.1. A vector space V over a field $\mathbb{F}$ is ...

1.1.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

2 Electrical Engineering Foundations

2.1 Electro Magnetism

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Sampling

2.3.2 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance